

New cross-cutting relationships: the four papers, each other, and the mainstream

This note answers a single question asked after the four drafts were finalised: *what do the findings, taken together, imply — and can we derive new, useful relationships between them and the existing literature?* Each relationship is **derived** here and then **verified one-by-one** in `glaciers/validation/synthetic/cross_relationships.py` (7 parametrised tests in `glaciers/tests/test_validation_synthetic.py`). No external data; CPU only.

The unifying thread of all four papers is one operator-theoretic statement:

A local, memoryless, down-gradient (eddy-diffusivity / “K-theory”) closure is the real-even-positive, Markovian-delta *projection* of an exact non-local operator. Each paper measures a different shadow that the projection throws away — a memory time, a migration, a backscatter, a fold.

The relationships below make that statement quantitative and tie it to mainstream results (Mori–Zwanzig/t-model, Kraichnan, Schoof MISI, Scheffer/Dakos early warning, Gilpin/Hanratty, Gudmundsson, rough Rayleigh–Bénard).

#	Relationship	Links	Status
NR1	One memory number $Me = \tau_{fast}/\tau_{slow}$ governs Markovian-closure breakdown	P1 $\rightleftharpoons$ P4 $\rightleftharpoons$ MZ/t-model	[VERIFIED]
NR2	K-theory = real-even-positive projection of one complex admittance Z	P3 $\rightleftharpoons$ P1 $\rightleftharpoons$ Kraichnan/Hanratty	[VERIFIED]
NR3	$s_N(N)$ flotation pole = Schoof MISI saddle-node; CSD scaling	P4 $\rightleftharpoons$ Schoof / Scheffer–Dakos	[VERIFIED]
NR4	$I = \tan\psi /2\pi$, ψ the Gilpin/Hanratty flux-topography phase	P3 $\rightleftharpoons$ Gilpin/Hanratty	[VERIFIED]
NR5	Melt ceiling = rough-RB no-enhancement branch (suppression $\uparrow$ steepness)	P2 $\rightleftharpoons$ rough-RB	[VERIFIED] (decomp.); crossover [<i>CONJECTURE</i>]
NR6	Tidal 2f/1f velocity ratio = curvature of the sliding law	P4 $\rightleftharpoons$ Gudmundsson	[VERIFIED]
NR7	Intrusion residence Ro is a Damköhler number	P4 $\rightleftharpoons$ transport theory	[DERIVED]

NR1 — One memory number governs every “local closure fails” verdict [**VERIFIED**]

Statement. Paper 1 shows single- K turbulence closure is the *Markovian-delta collapse* of a Mori–Zwanzig (MZ) memory kernel; Paper 4 shows the literal thermal sliding-lag kernel is wrong and the replacement is an *exact MZ memory kernel* $K(\tau) = M_{sq}M_{qs}e^{M_{qq}\tau}$ from eliminating the channel. These are the **same object** — the impedance Green’s function of an eliminated fast variable — and the validity of the *local* (Markovian) closure in both is set by **one** dimensionless number,

$$Me = \tau_{fast}/\tau_{slow} \text{ (eliminated-variable relaxation time / resolved timescale).}$$

Derivation. For the lumped pair $\dot{x} = Mx$, $M = [[-1/\tau_1, -a], [b, -1/\tau_2]]$, eliminating q gives the generalized Langevin equation $\dot{s} = M_{ss}s + \int K(t-\tau)sd\tau$ with $K(\tau) = -abe^{-\tau/\tau_2}$ and DC gain $\int K = M_{sq}M_{qs}/(-M_{qq})$. Holding the DC gain fixed (so the Markovian closure $\dot{s} = (M_{ss} + \int K)s$ is the correct *slow* limit), the residual is purely the finite-memory correction, which vanishes as $\tau_2 \rightarrow 0$. This is the Chorin–Hald–Kupferman / Stinis statement that the Markovian approximation improves as the timescales separate.

Verification. Sweeping $Me \in [0.02, 0.5]$, the L^∞ closure error rises monotonically $0.7\% \rightarrow 6.2\%$, vanishing as a positive power ($\approx Me^{0.69}$) — the same criterion in both systems. Paper 1’s measured $\tau_c \approx 0.02\text{--}0.03$ and Paper 4’s channel time τ_2 are the two instances. (nr1_memory_number.)

NR2 — K-theory is the real-even-positive projection of one complex admittance [VERIFIED]

Statement. A down-gradient closure is a **real, even, sign-definite** transport operator. Write the exact linear transport response to a sinusoid as one complex admittance $Z = a_r + ia_i$. Then:

- **Im $Z = a_i$** is the *quadrature* response: the scallop migration E_{cos} (Paper 3) **and** the closure memory phase (τ_c , Paper 1).
- **Re $Z < 0$** is *backscatter* — negative eddy viscosity (Paper 1, Kraichnan 1976).
- **K-theory = the projection** $Z \mapsto \max(\text{Re}Z, 0) + 0 \cdot i$. It zeroes the migration **and** the backscatter at once.

Derivation. For topography $y = a \sin(kx)$ the flux $e = \text{Re}[Z \cdot \hat{y} \cdot e^{ikx}]$ with $\hat{y} = -ia$ gives $e = a[a_r \sin(kx) + a_i \cos(kx)]$, so $E_{sin} = aa_r$ (amplitude rate, Re s) and $E_{cos} = aa_i$ (migration, Im s). A real diffusive flux of a sinusoid is in-phase $\rightarrow E_{cos} = 0$ (Paper 3’s keystone). The transfer sign $T \sim -\text{Re}Z$ is backscatter iff $a_r < 0$ (Kraichnan’s 2-D negative eddy viscosity).

Verification. With $Z = -1 - 0.6i$: exact $E_{sin} = -1$, $E_{cos} = -0.6$, migration $I = 0.0955$, backscatter present; the K-theory projection gives $E_{cos} = 0$, $I = 0$, no backscatter. The migration and the backscatter are independent components that vanish *together* only for a genuine down-gradient closure. (nr2_admittance_unification.)

NR3 — The s_N flotation pole is the Schoof MISI saddle-node [VERIFIED]

Statement. Paper 4’s effective-pressure master curve $|s_N|(N) = m/(1 - (N_c/N)^m)$ has a simple pole at the flotation threshold N_c . That pole is the loss of basal-drag restoring stiffness — i.e. the **Schoof (2007) grounding-line saddle-node** that underlies marine ice-sheet instability — and the velocity critical-slowness early-warning (Paper 4) is the **Scheffer (2009) / Dakos (2008)** generic precursor of that fold.

Derivation. Near $N = N_c(1 + \delta)$, $R = (N_c/N)^m \approx 1 - m\delta$, so the sliding sensitivity $|s_N| = m/(1 - R) \propto \delta^{-1}$ (the pole), while the restoring stiffness $d\tau_b/du \propto (1 - R)^2/R \propto \delta^{+2}$. The restoring eigenvalue $\lambda \propto \delta^2 \rightarrow 0$ at the fold; by the standard CSD result $\text{Var} \propto 1/\lambda \propto \delta^{-2}$ and $AC1 = e^{-\lambda\Delta t} \rightarrow 1$. Schoof’s flux $q_g \propto h_g^{(m+n+3)/(m+1)} \approx h_g^{4.75}$ blows up at the same flotation thickness.

Verification. Fitted near-fold exponents: $|s_N|$: -0.997 , stiffness $+1.998$, variance -1.998 ; $AC1$: $0.905(\text{far}) \rightarrow 0.9999(\text{near})$. An OU realization with N declining toward N_c gives rising rolling variance and $AC1$ (Kendall $\tau = 0.54, 0.55$) — the operational early-warning. (nr3_misi_fold_ews.)

NR4 — The constant-free field ratio is the Gilpin/Hanratty phase [VERIFIED]

Statement. Paper 3’s ΔT -free, constant-free field test $I = Im(s)/(2\pi|Re(s)|)$ is exactly

$$I = |\tan\psi|/(2\pi), \psi = \arg(s) = \text{atan2}(E_{cos}, E_{sin}),$$

where ψ is the **Gilpin–Hirata–Cheng (1980) / Hanratty (1981)** heat-flux-to-topography phase shift. Downstream migration $\Leftrightarrow \psi \in (\pi/2, \pi)$ (Gilpin’s criterion) $\Leftrightarrow$ damped ($Res < 0$) and migrating ($Im s \neq 0$). The K-theory limit is $\psi \rightarrow 0$ ($I \rightarrow 0$).

Verification. On Paper 3’s own measured (E_{sin}, E_{cos}) table the two expressions for I agree to machine precision (0.168, 0.197, 0.248, 0.394), and every point has $|\psi| \in (90^\circ, 180^\circ)$ ($-133^\circ \dots -112^\circ$) — the damped-downstream branch. (nr4_migration_phase.)

NR5 — The melt ceiling is the rough-RB no-enhancement branch [VERIFIED] decomposition

Statement. Paper 2’s melt ceiling $Nu/Nu_{flat} \leq 1$ is the **no-enhancement** branch of rough Rayleigh–Bénard heat transfer. Writing $Nu/Nu_{flat} = A \cdot f_{loc}$ with A the wetted-area gain and f_{loc} the local-flux factor, the ceiling *proves* the cold wall suppresses local flux faster than the geometry adds area.

Derivation. For $y = a \sin(kx)$, $A = \langle \sqrt{1+y'^2} \rangle \approx 1 + \pi^2(a/\lambda)^2 > 1$ always. Since Paper 2 measures $Nu/Nu_{flat} \approx 0.96\text{--}0.97 \leq 1$, the implied $f_{loc} = (Nu/Nu_{flat})/A < 1$ and the suppression $1 - f_{loc}$ must **rise** with steepness — exactly cancelling the area gain. Enhancement ($Nu/Nu_{flat} > 1$) requires $f_{loc} > 1/A$, i.e. the Paper-2 §G.6 lee-flux growth term to overcome the deficit — a regime crossover flagged [CONJECTURE] (it needs the lee/separation data to place).

Verification. Across $a/\lambda \in [0.05, 0.20]$: $A : 1.02 \rightarrow 1.39$, $Nu/Nu_{flat} : 0.97 \rightarrow 0.96$, implied $f_{loc} : 0.95 \rightarrow 0.69$, suppression $0.05 \rightarrow 0.31$ (monotone rising). (nr5_melt_ceiling_suppression.)

NR6 — The tidal 2f/1f ratio measures the curvature of the sliding law [VERIFIED]

Statement. Under a tide $N(t) = N_0(1 + \varepsilon \cos \omega t)$, the surface-velocity second-harmonic ratio is

$$2f/1f = (\varepsilon/4) \cdot |s'_N/s_N - 1 + s_N|, \quad s'_N = ds_N/d\ln N,$$

i.e. it reads the **curvature** of the sliding law and diverges toward N_c . This is the sliding-law origin of Gudmundsson’s observed nonlinear (MSf) tidal response, and answers the “no knowledge of N ” objection (it is recovered from velocity alone).

Derivation. Taylor-expanding $\ln u_b(\ln N)$ to second order and exponentiating (velocity, not log-velocity, is the observable — hence the $+s_N$ from *exp*) gives the fundamental amplitude $\propto s_N \varepsilon$ and the second harmonic $\propto (\varepsilon^2/4)(s'_N - s_N + s_N^2)$.

Verification. Synthetic tide + FFT at $N_0/N_c = 20, 8, 3, 1.5$: the measured 2f/1f rises monotonically toward flotation and matches the closed-form to $\leq 15\%$. (nr6_tidal_curvature.)

NR7 — The intrusion residence number is a Damköhler number

[DERIVED]

Statement. Paper 4's $Ro = v_{kin} \cdot \tau_{hyd}/\ell$ (with $v_{kin} = A \cdot dH/dt$) is a **Damköhler number**: the ratio of the kinematic intrusion-front advance to the hydraulic relaxation. $Ro = 1$ at $\tau_{crit} = \ell/v_{kin}$ separates thinning-paced ($Ro < 1$) from hydraulic-limited ($Ro > 1$) intrusion — the transport-vs-process crossover familiar from reaction-transport theory.

Verification. With Paper-4 runaway-tail numbers ($v_{kin} = 1.05 \text{ km/yr}$, $\ell = 1 \text{ km}$): $\tau_{crit} = 0.95 \text{ yr}$; sweeping τ_{hyd} gives $Ro = 0.011, 0.105, 1.05, 2.1$, flipping regime at $Ro = 1$. (nr7_residence_damkohler.)

What is genuinely new vs. a sharpening

- **New unifications:** NR1 (one memory number across turbulence *and* sliding), NR2 (one complex admittance whose two parts are Paper 3 migration and Paper 1 backscatter), NR3 (the s_N pole *is* the Schoof MISI saddle-node, with the CSD exponents derived).
- **Sharpenings of connections the papers already gesture at:** NR4 (the explicit $I = |\tan\psi|/2\pi$ identity to Gilpin/Hanratty), NR6 (the closed-form tidal curvature ratio), NR5 (the area/flux decomposition of the melt ceiling), NR7 (naming Ro a Damköhler number).

Honest limits

NR1's error exponent is metric-dependent (≈ 0.69 in L_∞); the claim is the *monotone vanishing*, not a universal power. NR2's admittance is a linear, single-mode model of the discarded structure, not a turbulence closure. NR3 identifies the bifurcation and the local EWS scaling; it does not run a full grounding-line model. NR5's enhancement crossover is a conjecture pending lee-flux data. NR6 is single-mechanism (tidal N modulation only). All references remain [cite] slots — none invented.